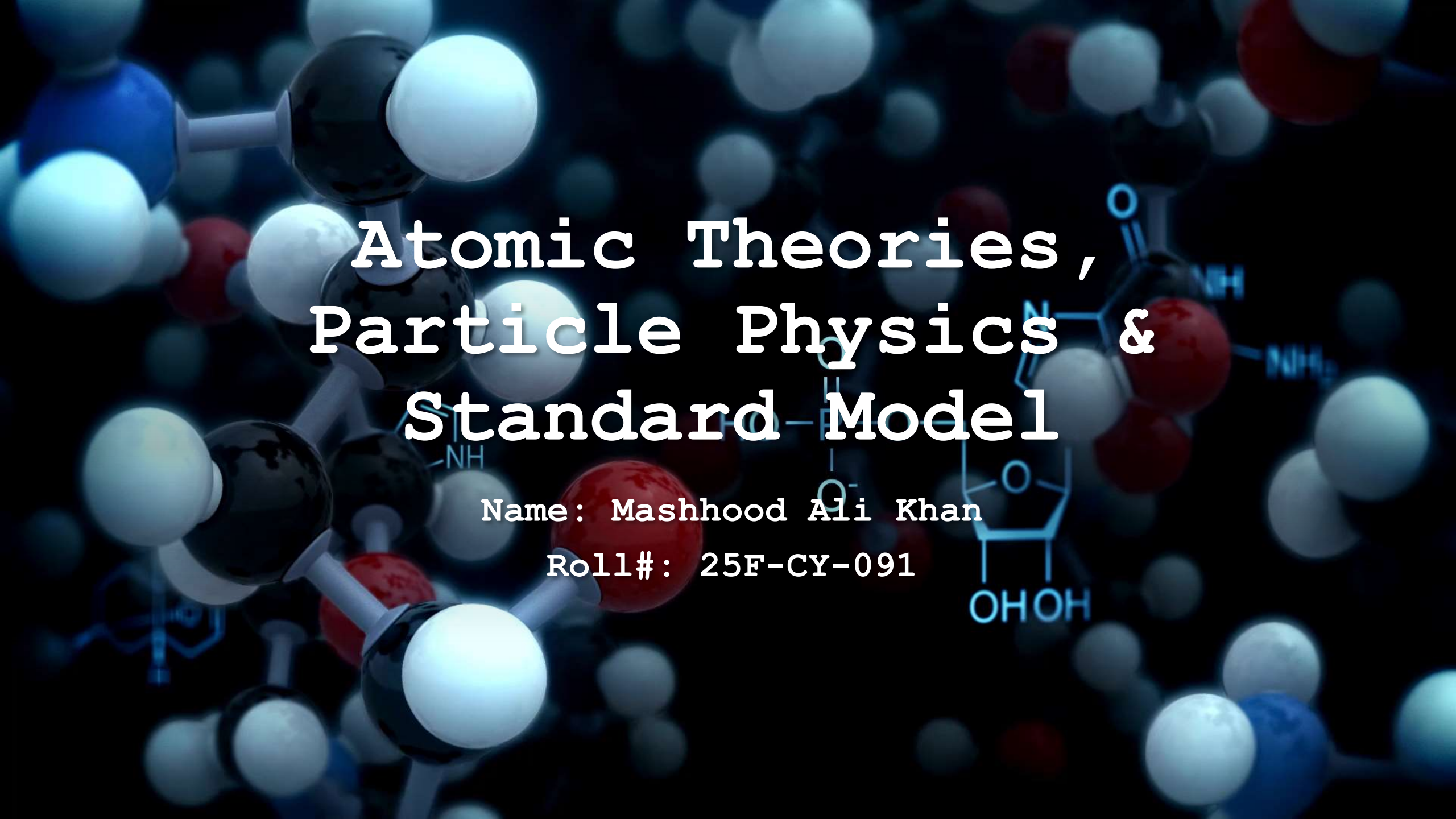




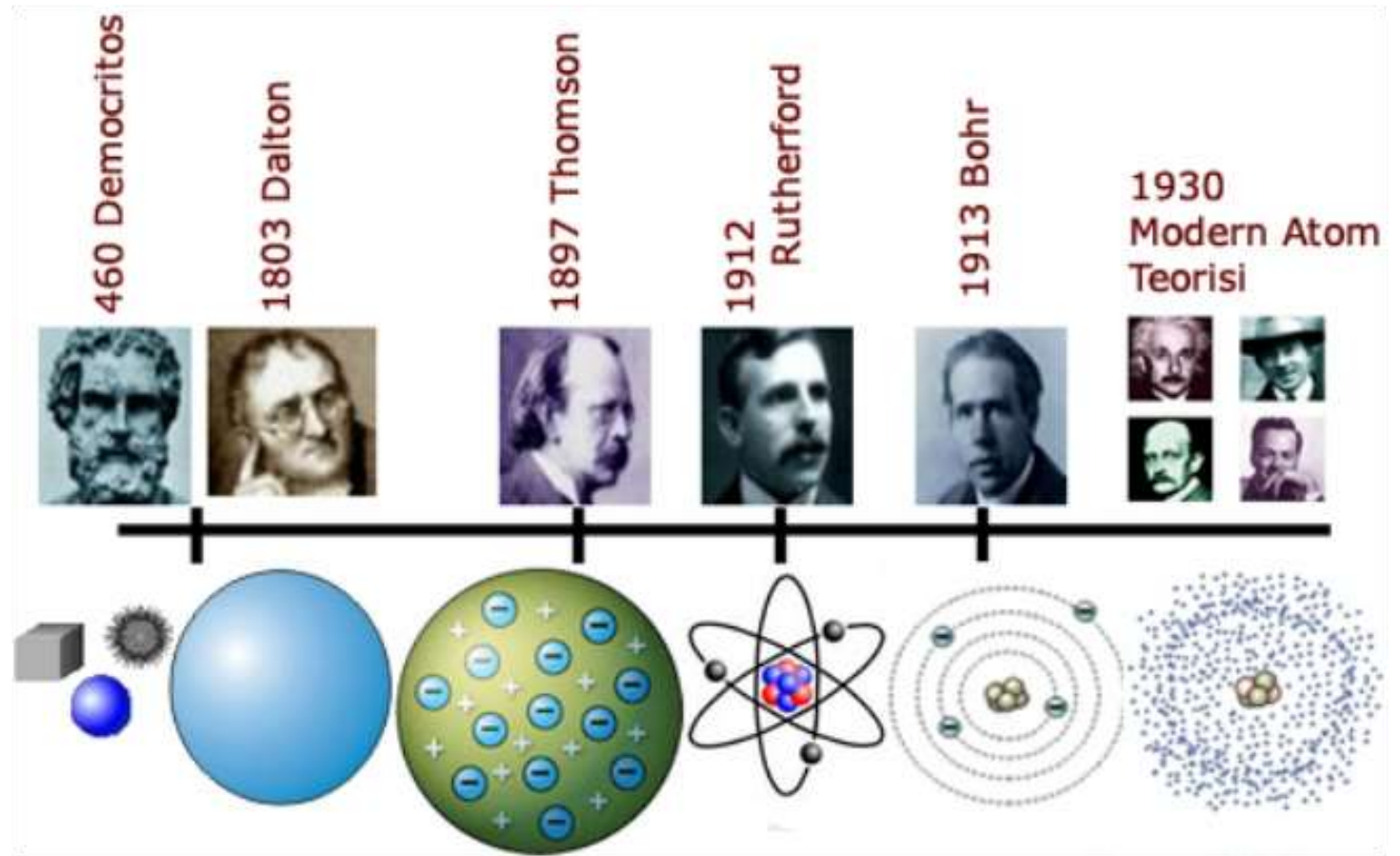
Atomic Theories, Particle Physics & Standard Model

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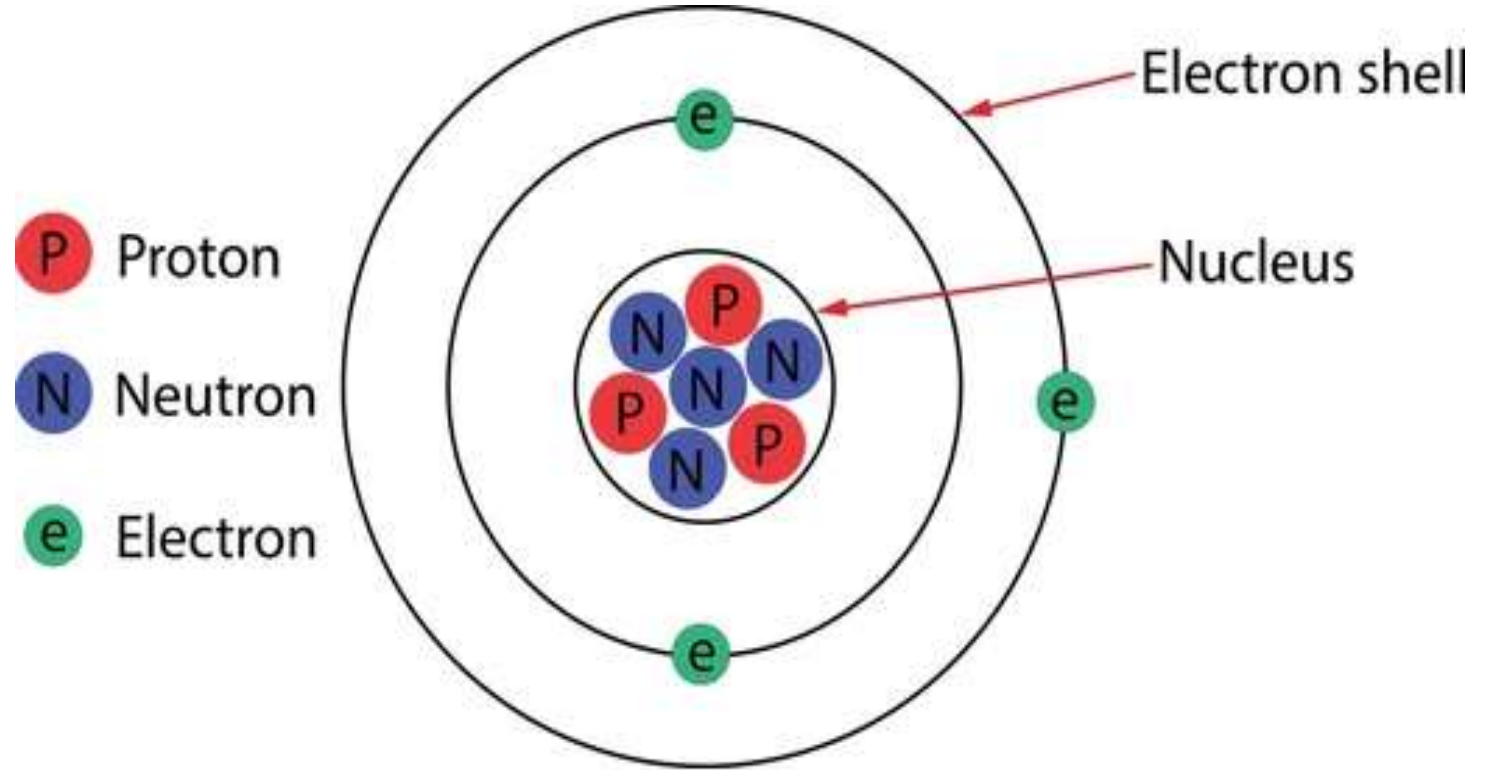
Introduction to Atomic Theory

- Atomic theory explains the structure and behavior of matter
- It states that all matter is made up of very small particles called atoms
- Atomic theory developed through experiments over many years
- It forms the basis of modern physics



What is an Atom?

- An atom is the smallest unit of matter
- It retains the chemical properties of an element
- An atom consists of:
 - Nucleus (protons + neutrons)
 - Electrons revolving around the nucleus
- Atoms combine to form molecules



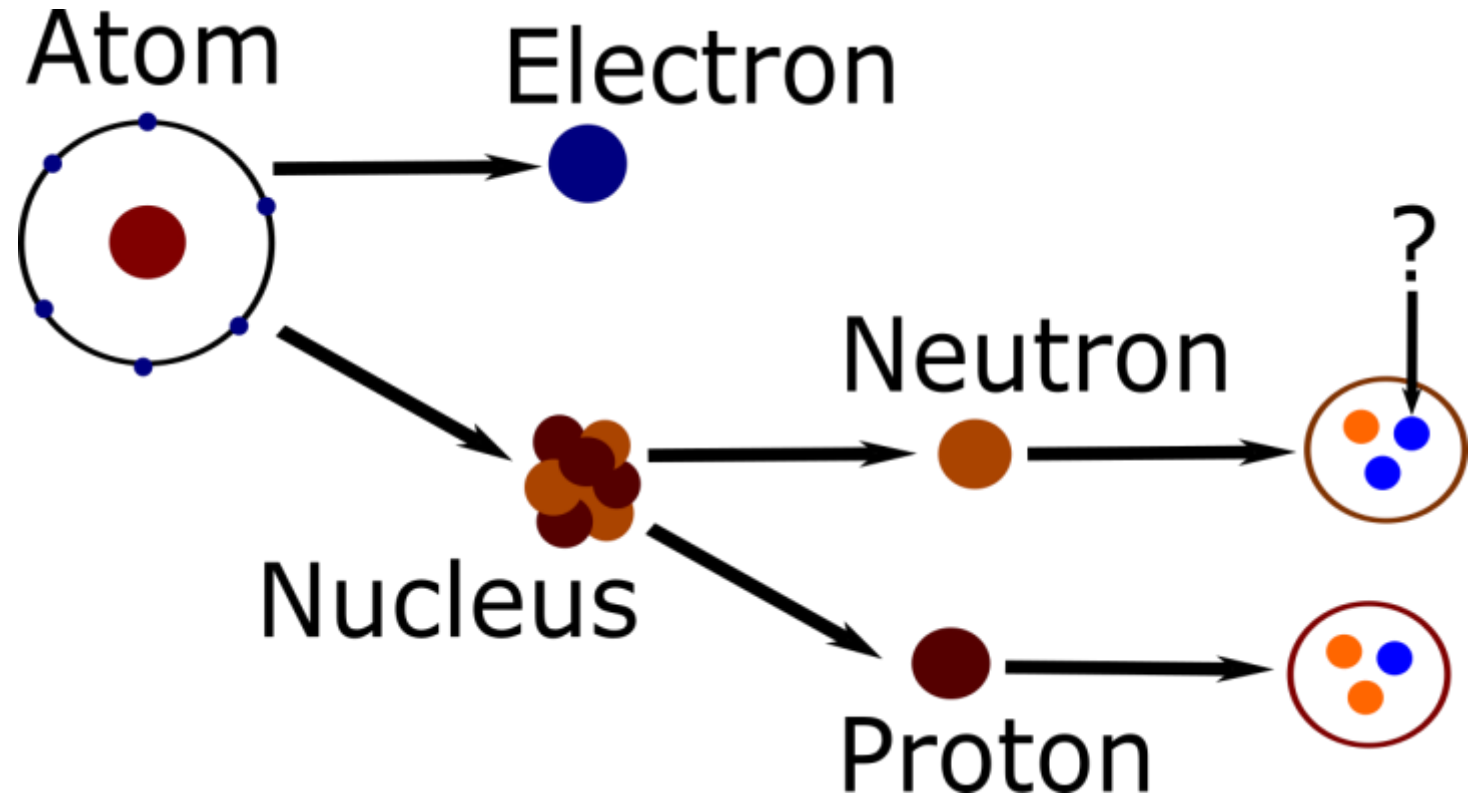
Atomic Particles

- **Protons:** positively charged particles

- **Neutrons:** neutral particles

- **Electrons:** negatively charged particles

- Protons and neutrons determine atomic mass



Dalton's Atomic Theory (Introduction)

Proposed by
John Dalton in
1803

First modern
atomic theory

Explained laws
of chemical
reactions

Dalton's Atomic Theory (Postulates)

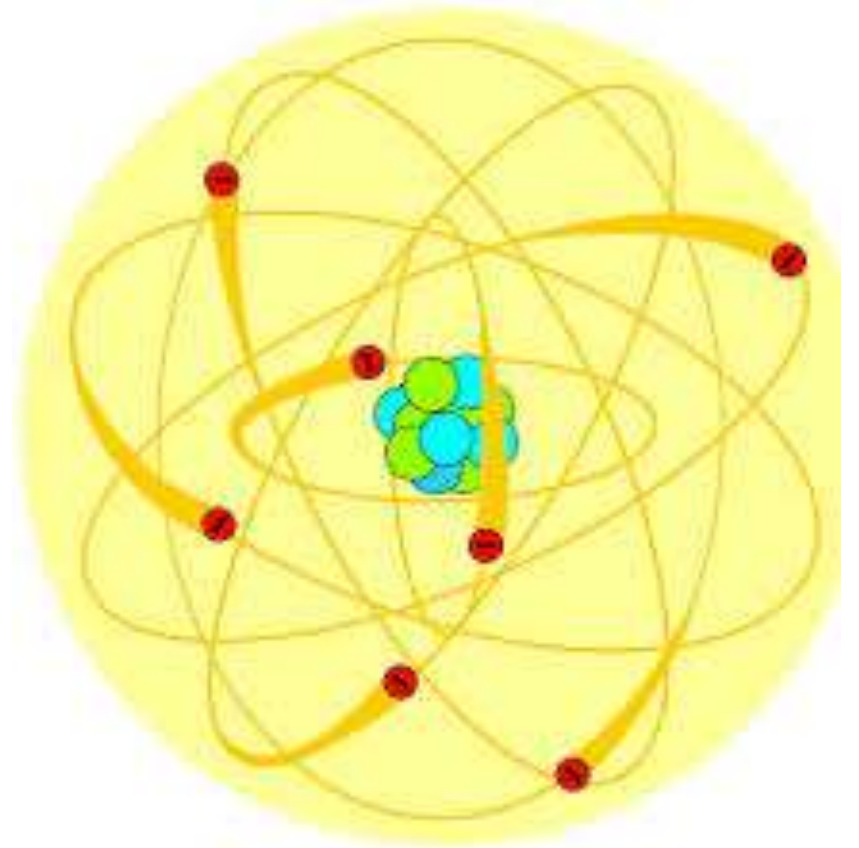
All matter is composed of atoms

Atoms of the same element are identical

Atoms of different elements are different

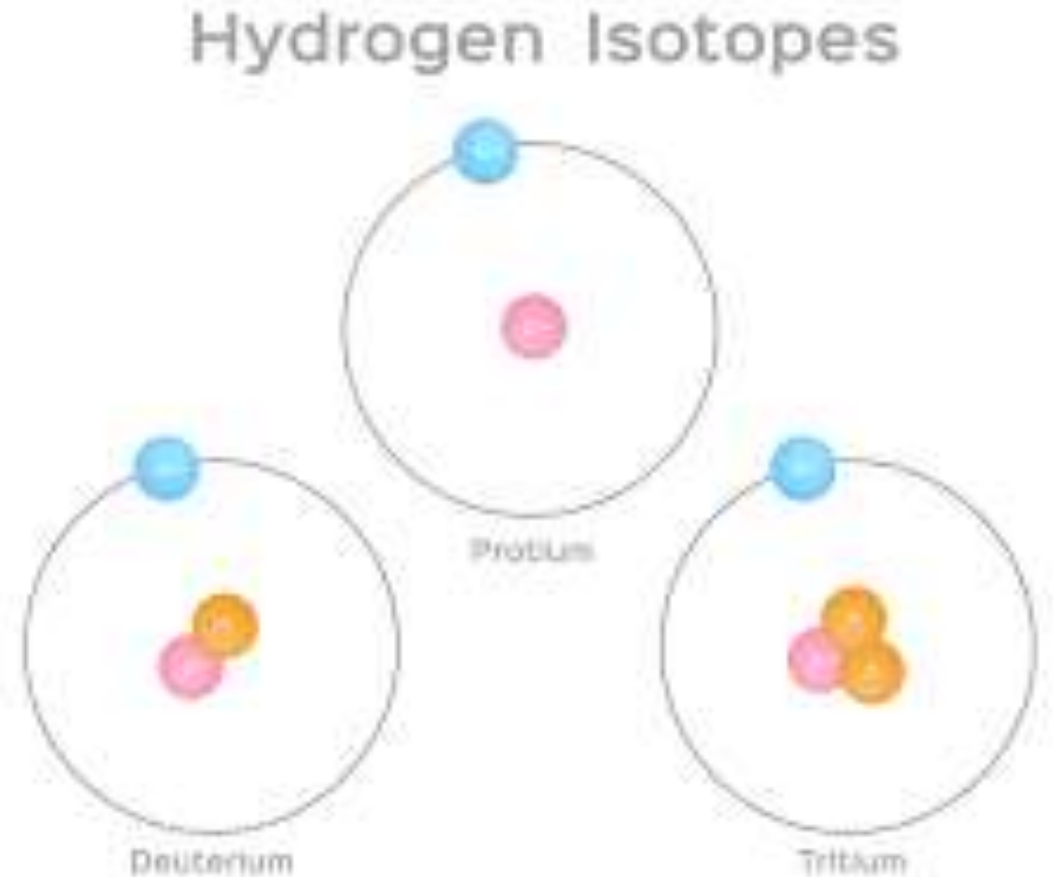
Atoms combine in simple whole-number ratios

Atoms are neither created nor destroyed



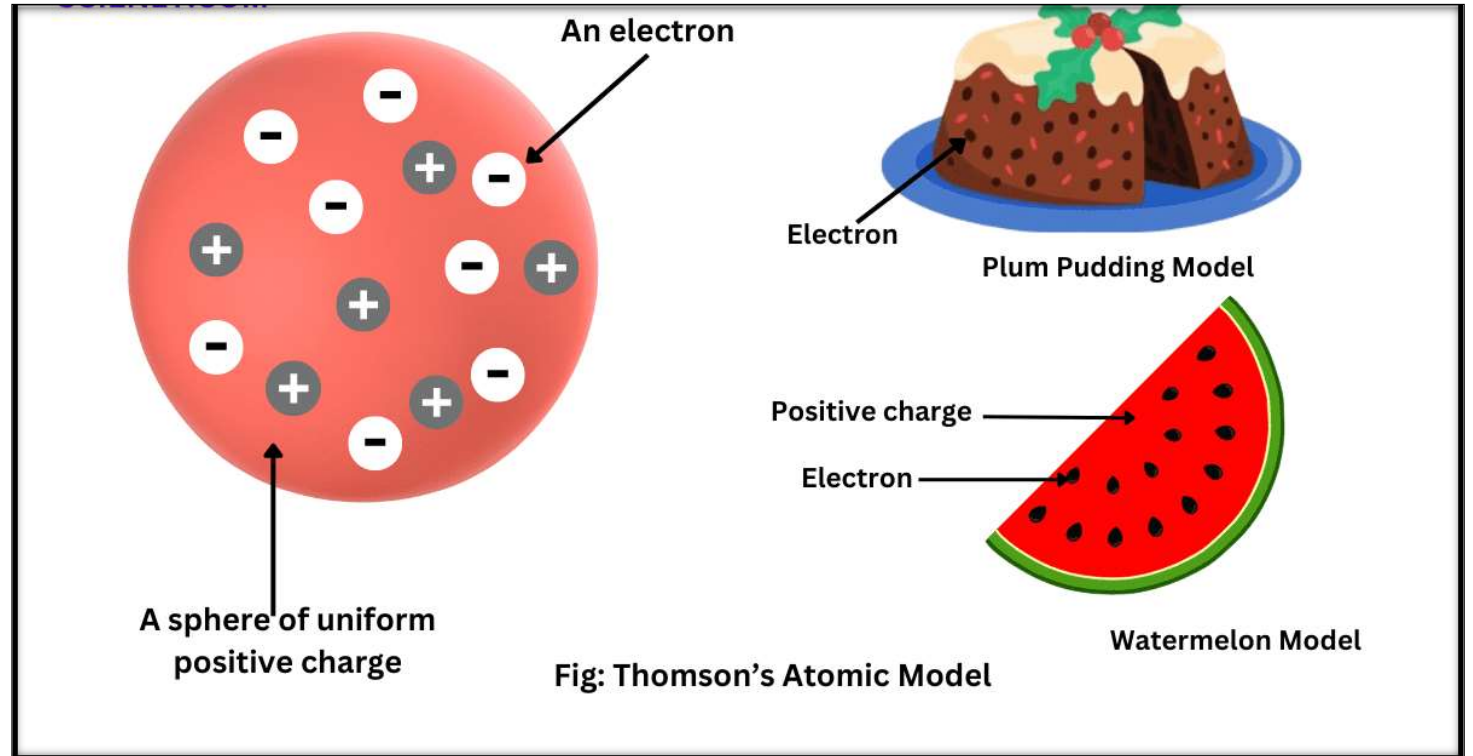
Limitations of Dalton's Theory

- Atoms are divisible (discovery of subatomic particles)
- Atoms of same element are not always identical (isotopes)
- Does not explain internal structure of atom



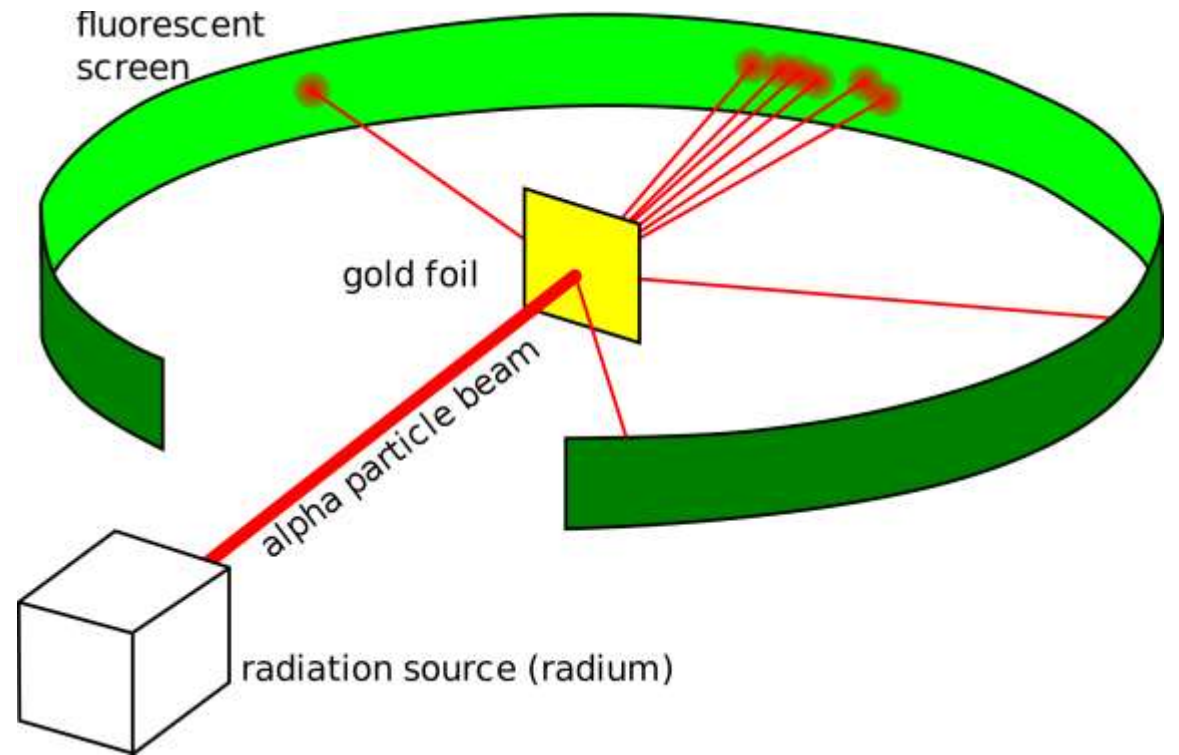
Thomson's Atomic Model

- Proposed by J.J. Thomson in 1904
- Discovered the electron using cathode ray experiment
- Atom is a positively charged sphere
- Electrons are embedded inside
- Known as **Plum Pudding Model**



Rutherford's Atomic Model

- Proposed by Ernest Rutherford (1911)
- Based on gold foil experiment
- Most alpha particles passed through foil
- Few particles were deflected



Rutherford's Conclusions

Atom has a small,
dense nucleus

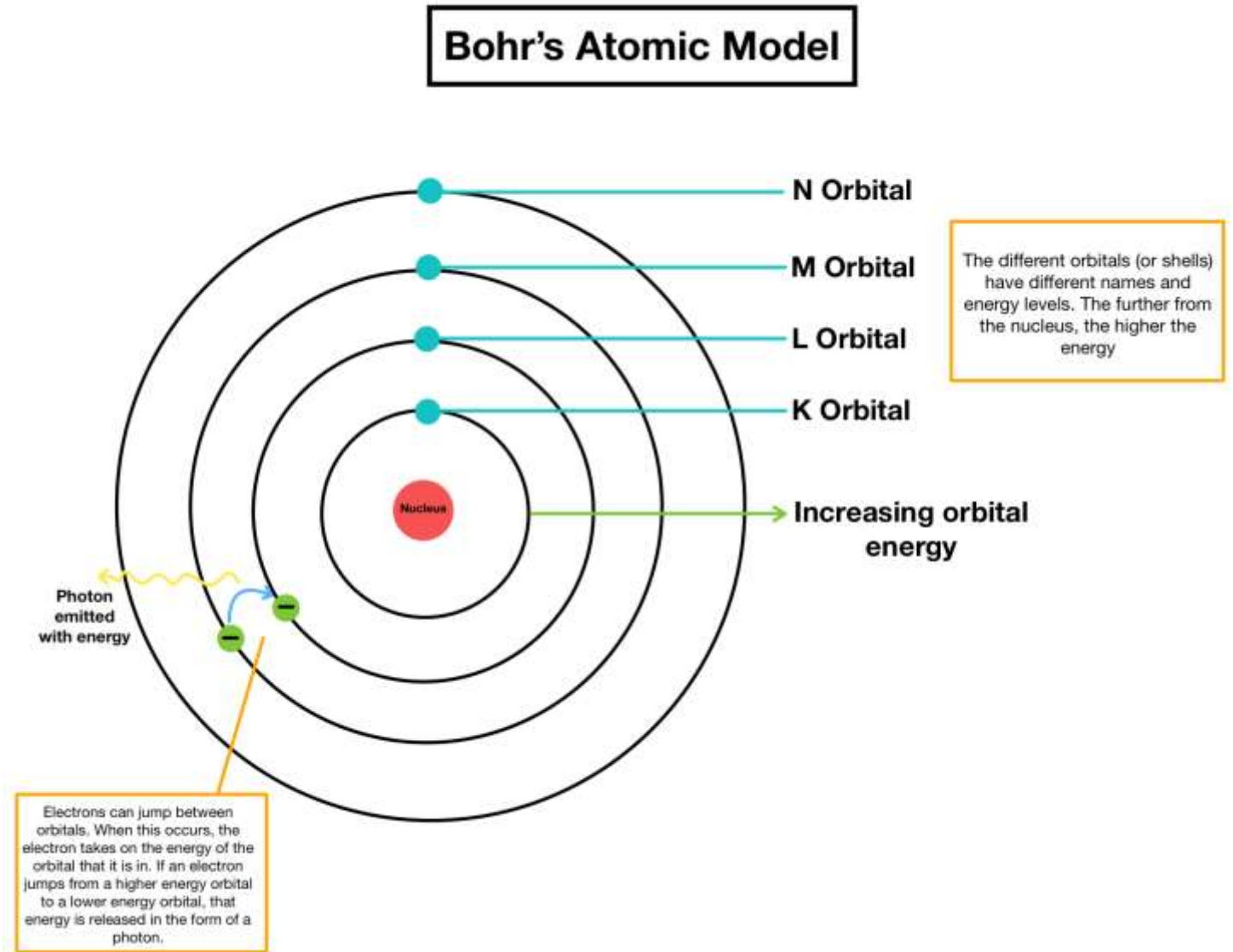
Nucleus is positively
charged

Most of the atom is
empty space

Electrons revolve
around nucleus

Bohr's Atomic Model

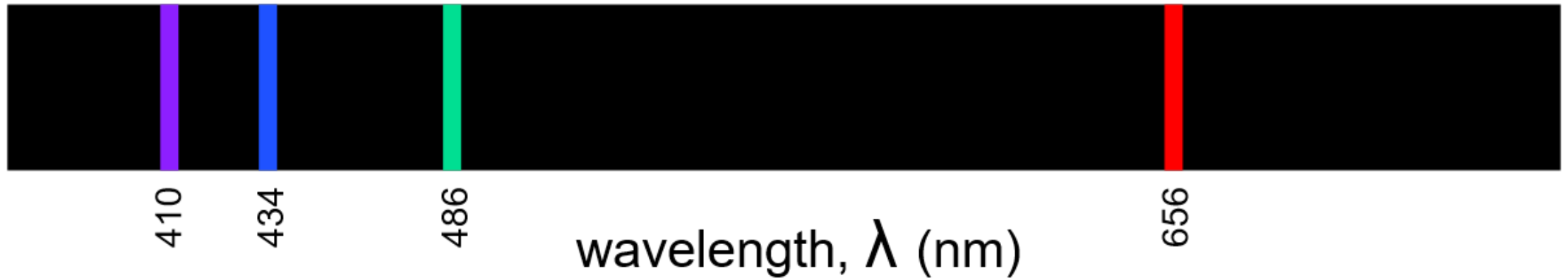
- Proposed by Niels Bohr (1913)
- Electrons move in fixed circular orbits
- Each orbit has fixed energy
- Energy is absorbed or emitted during transitions



Importance of Bohr's Model

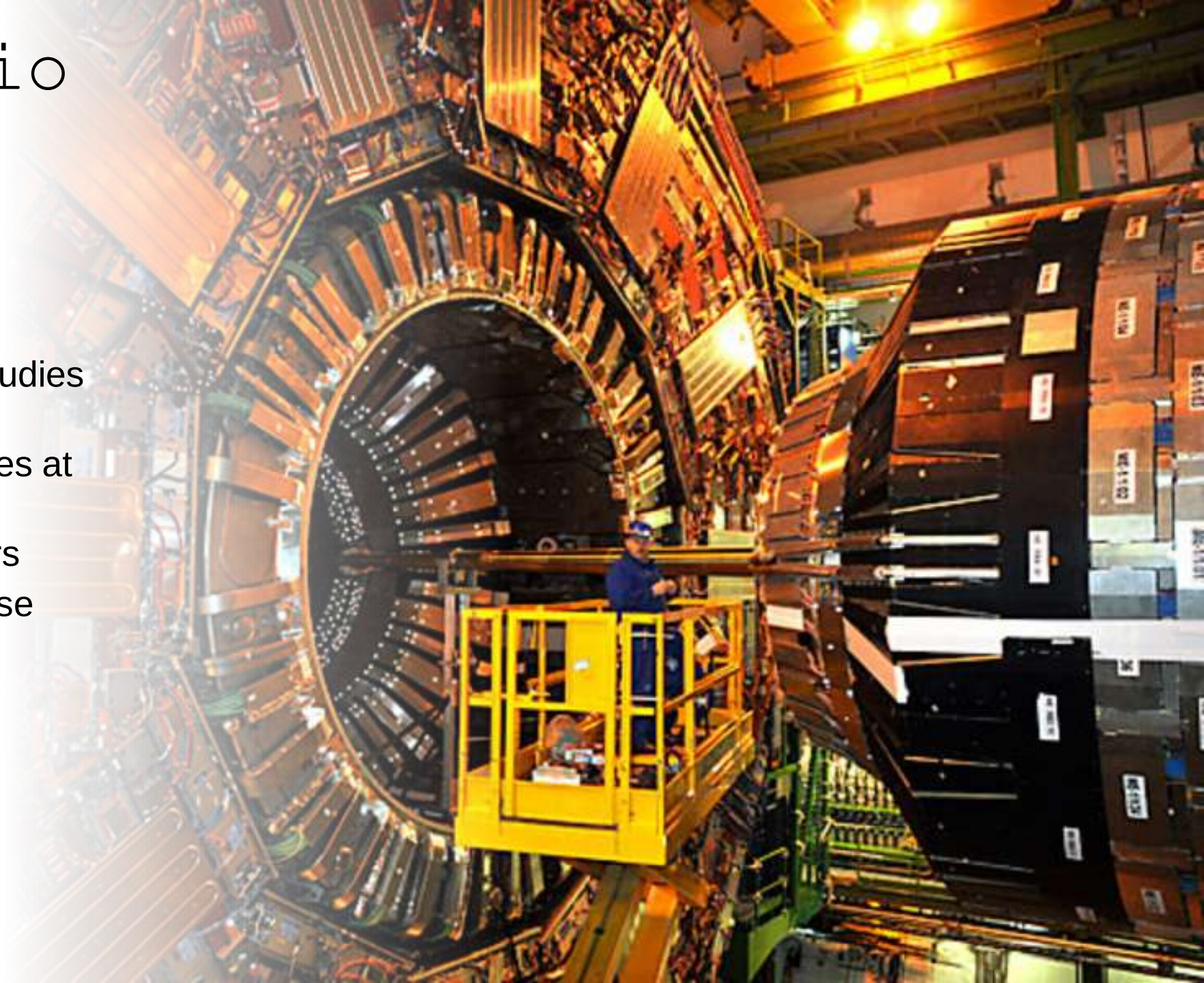
- Explained stability of atom
- Explained hydrogen spectrum
- Introduced energy levels

Hydrogen Emission Spectrum



Introduction to Particle Physics

- Branch of physics that studies fundamental particles
- Explains matter and forces at smallest scale
- Uses particle accelerators
- Helps understand universe structure

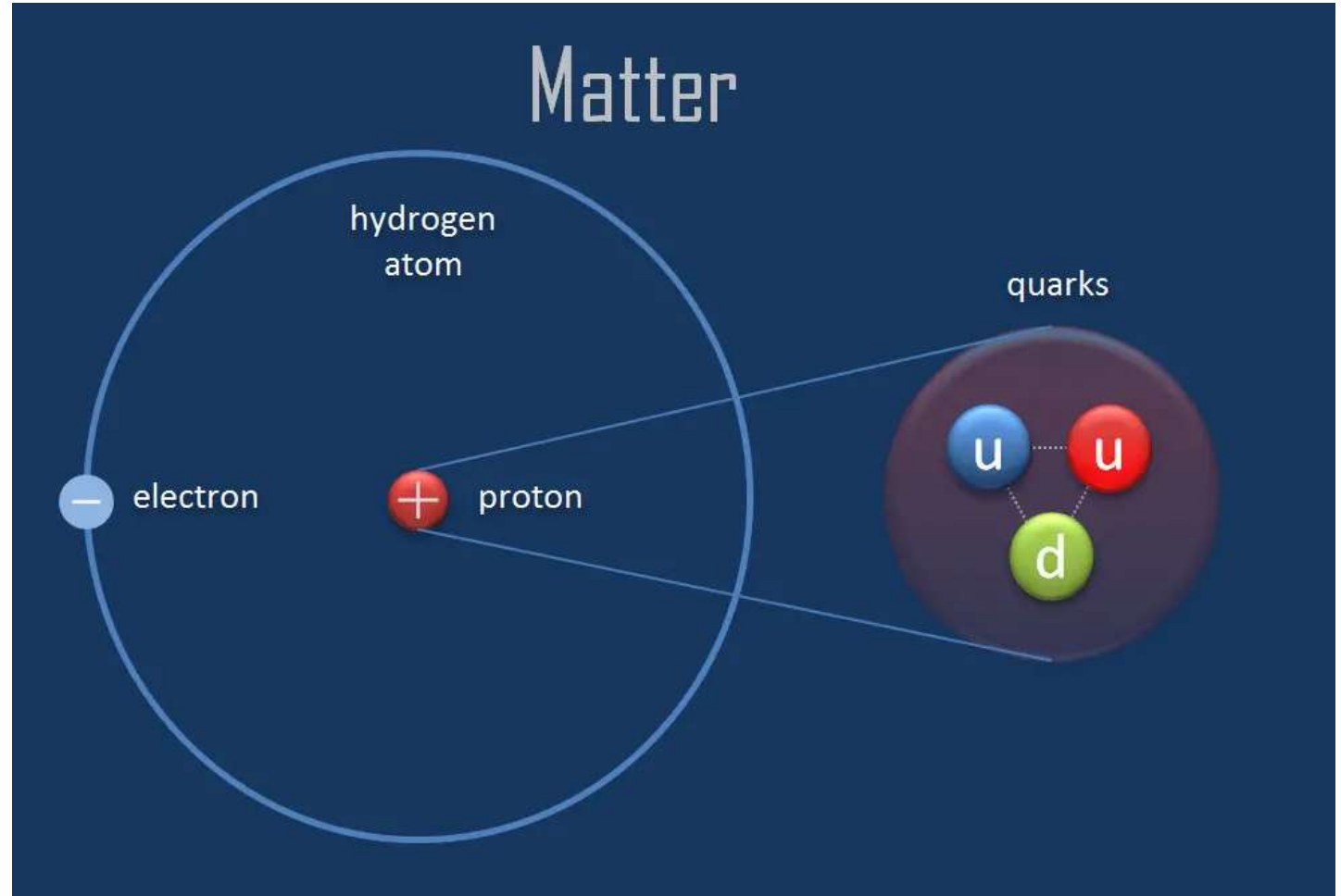


Fundamen tal Particle s

- Fundamental particles have no internal structure
- Two main categories:
- Fermions
- Bosons

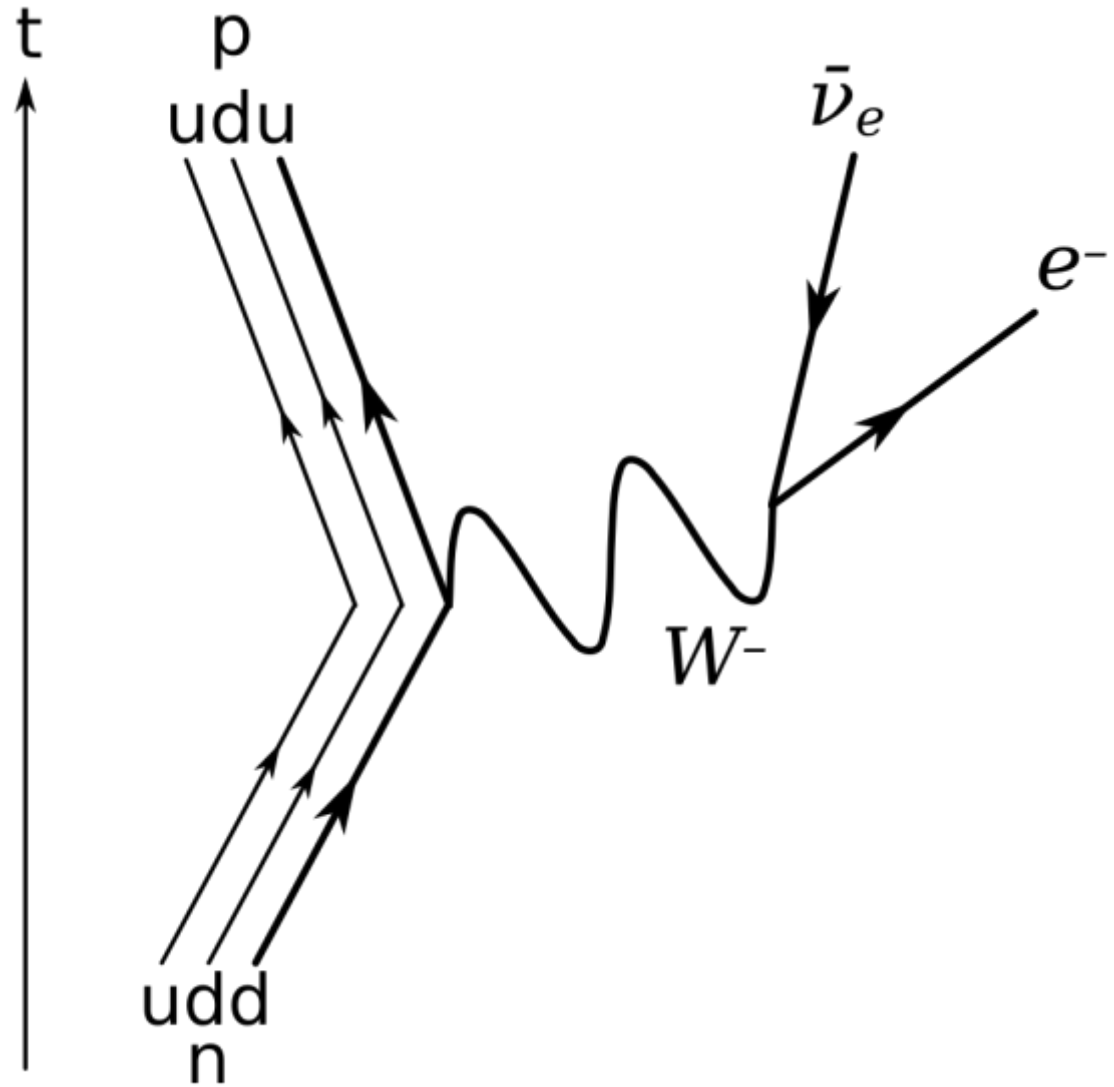
Quarks

- Building blocks of protons and neutrons
- Six types: up, down, charm, strange, top, bottom
- Experience strong nuclear force
- Cannot exist independently



Leptons

- Do not experience strong nuclear force
- Include electron, muon, tau
- Each has a neutrino
- Electron is most stable lepton



Bosons

- Force-carrying particles
- Photon → electromagnetic force
- Gluon → strong force
- W and Z bosons → weak force
- Higgs boson → mass

Standard
Model
(Introduction)

Theory that
explains
fundamental
particles three
fundamental
forces
One of the most
successful
theories

Forces in the Standard Model

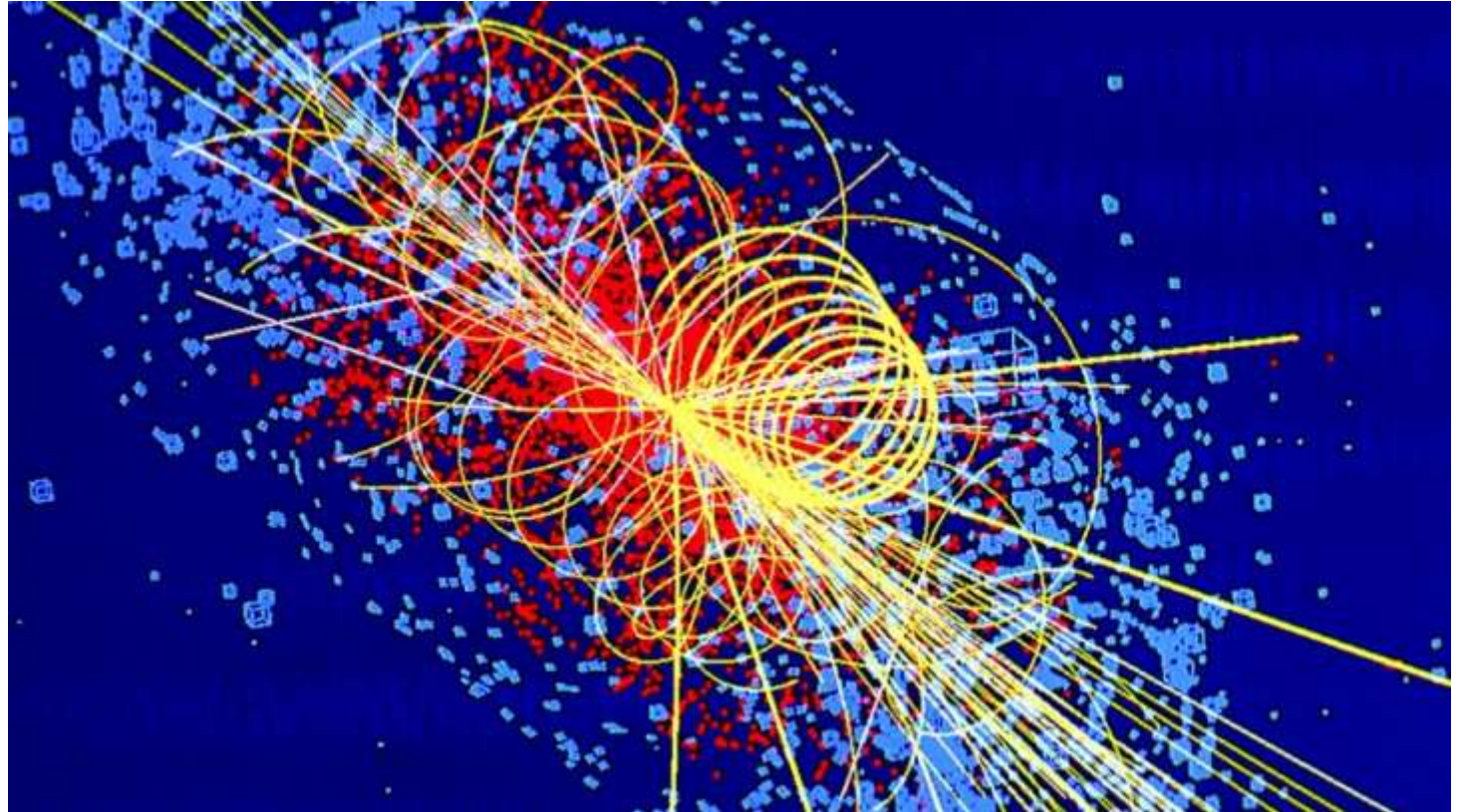
Electromagnetic
force

Strong nuclear
force

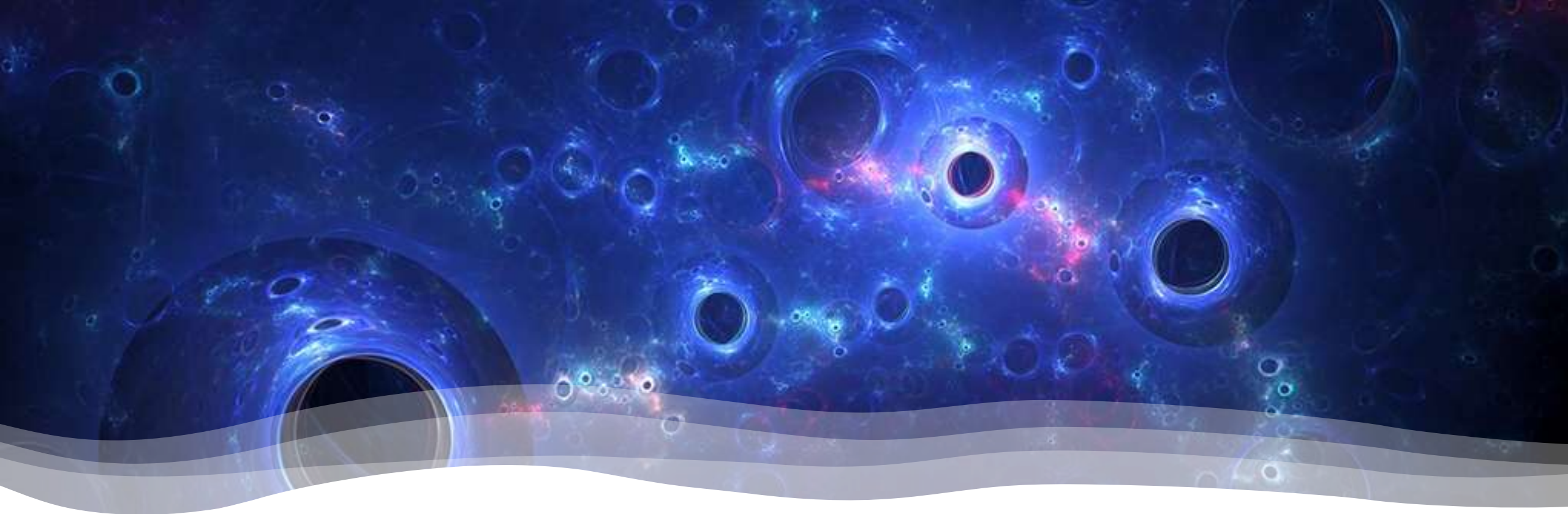
Weak nuclear
force

Gravity is not
included

Higgs Boson



- Associated with Higgs field
- Gives mass to particles
- Predicted in 1964
- Discovered in 2012 at CERN



Limitations of Standard Model

- Does not explain gravity
- Cannot explain dark matter
- Needs further development

Conclusion

- Atomic theories evolved with experiments
- Particle physics explores deeper structure of matter
- Standard Model explains fundamental particles and forces
- Research in physics is still ongoing

